

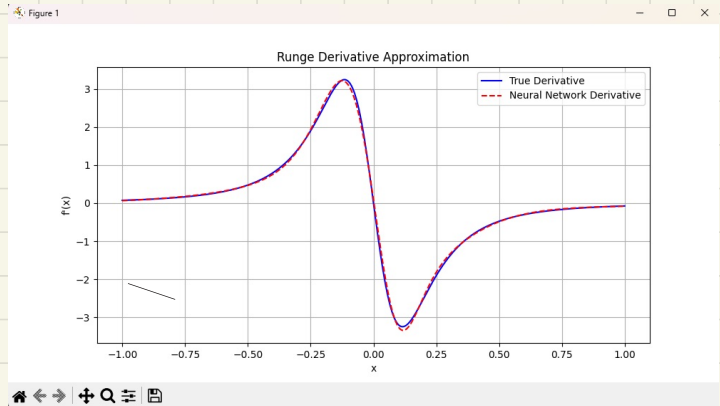
Assignment 3 - Programming

(1) Method : same as assignment 3

$$\text{Derivative MSE} : \frac{1}{n} \sum \|f'(x_i) - \hat{f}'(x_i)\|^2$$

Derivative MSE: 0.002417

Derivative Max Error: 0.171086



(2) Method

data : random to get 1000 point in $[-1, 1]$

hypothesis : $\hookrightarrow$ 80% train 20% validation

1, activation function = $\tanh x$

2, one input neuron

two hidden layers with 20 neuron

two output neuron (for $f(x)$ & $f'(x)$)

loss :

$$\text{function MSE} = \frac{1}{n} \sum (f(x_i) - \hat{f}(x_i))^2$$

$$\text{derivative MSE} = \frac{1}{n} \sum (f'(x_i) - \hat{f}'(x_i))^2$$

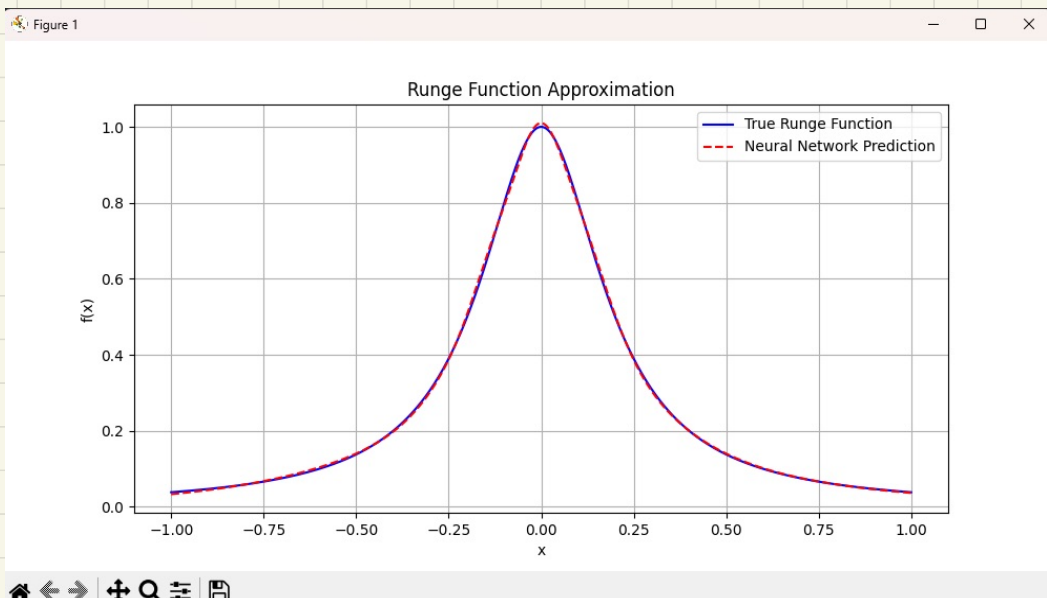
$$\text{LOSS} = \frac{1}{2} \text{MSE}_{\text{func}} + \frac{1}{2} \text{MSE}_{\text{deriv}}$$

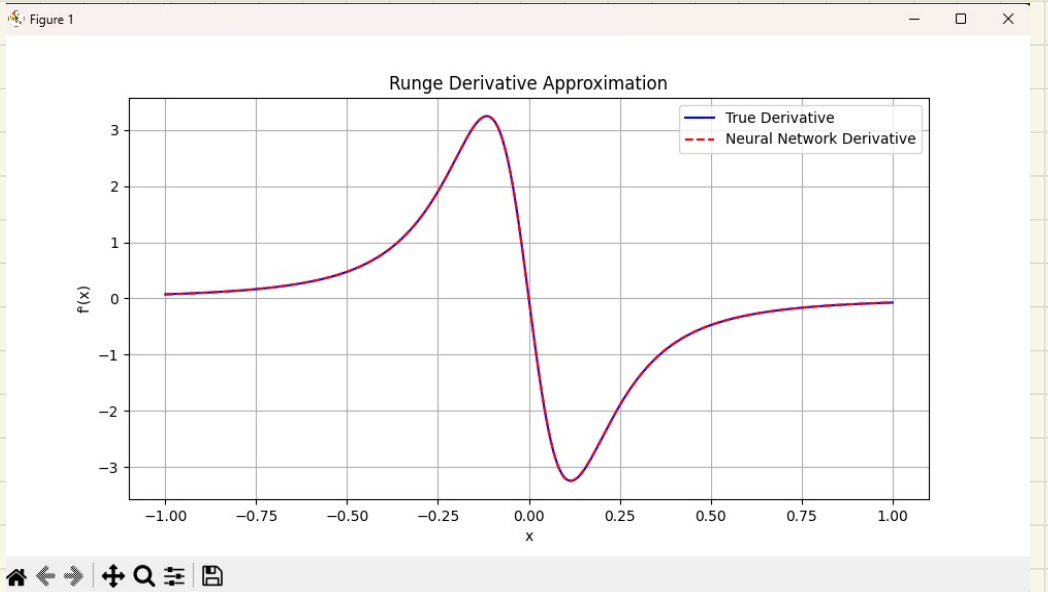
Optimizer :

the network trains for 1000 epochs .

Result

```
Epoch [1000/1000], Train Loss: 0.000021  
Function MSE: 0.000021  
Function Max Error: 0.014412  
Derivative MSE: 0.000010  
Derivative Max Error: 0.009954
```





Discussion :

both $\tilde{f}(x)$ & $\hat{f}'(x)$ look like $f(x)$ and $f'(x)$
I think it's successful.